

Lecture #183b

OMM: Trigger Points

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Objectives

1. Understand the criteria for diagnosis of trigger points and the underlying pathology
2. Review specific trigger points and referral patterns
3. Understand the differences between trigger points and counterstrain points, fibromyalgia
4. Explain the treatment process as well as prevention of trigger points, including stretching and postural considerations

What are Trigger Points?

Travell Definition:

“A trigger point is a hyperirritable spot in a taut band of a skeletal muscle that is painful on compression, stretch, overload or contraction of the tissue which usually responds with a referred pain that is perceived distant from the spot.”

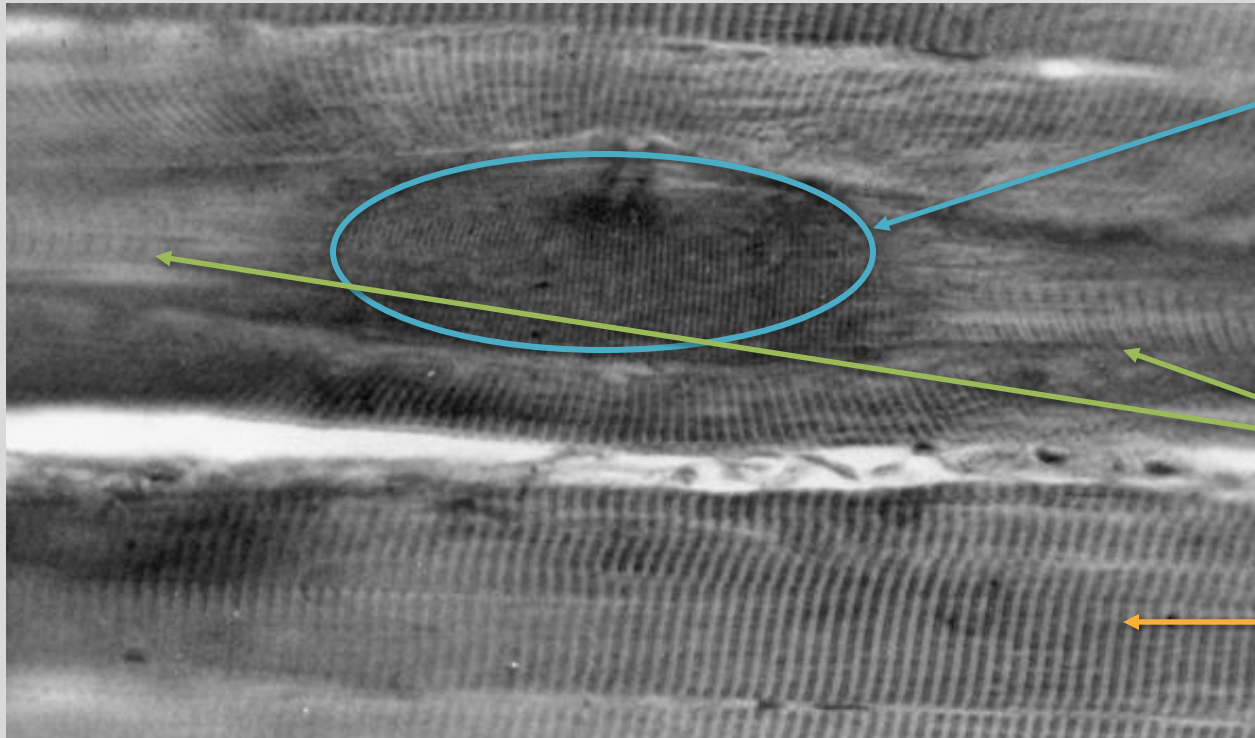
Component of Myofascial Pain Syndrome.

Characteristics

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Biopsy of a “taut band” in a canine gracilis muscle



“Knot” in taut band with severe contracture of sarcomeres

Compensatory elongation of sarcomeres

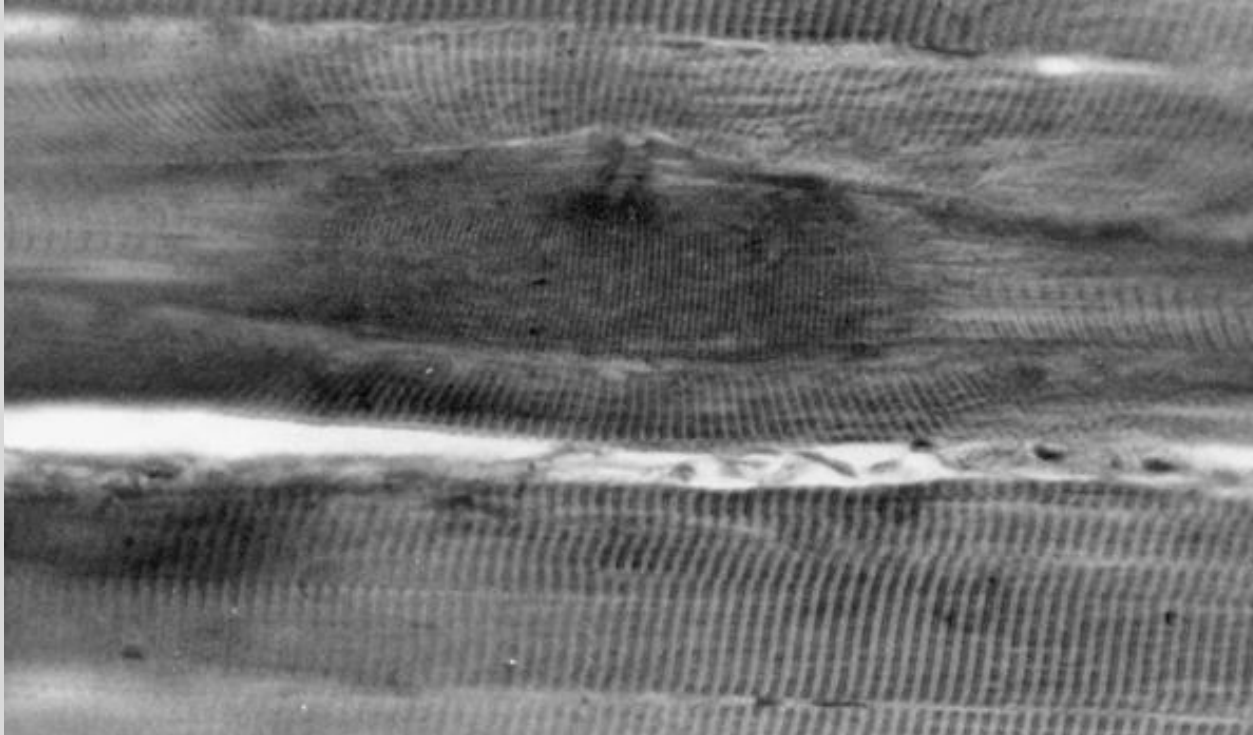
Normal muscle with regularly-spaced sarcomeres

Characteristics

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Biopsy of a “taut band” in a canine gracilis muscle
Striations correspond to sarcomere length



“Knot” in taut
band with severe
contracture of
sarcomeres

Compensatory
elongation of
sarcomeres

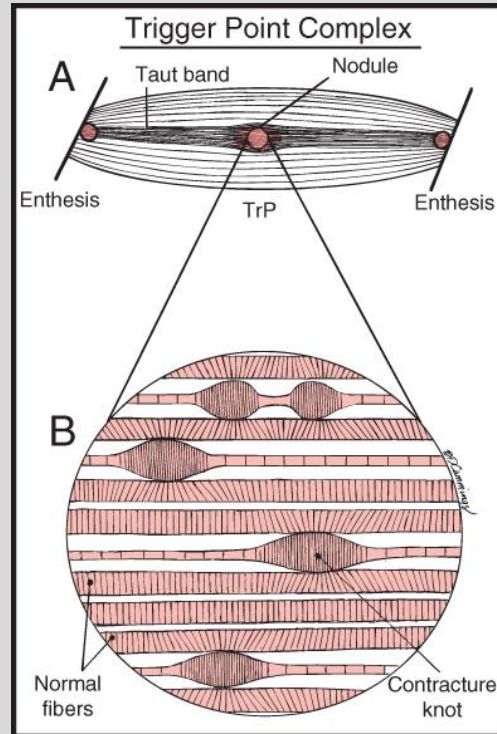
Normal muscle
with regularly-
spaced sarcomeres

Characteristics

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Trigger point complex in a longitudinal section of muscle:



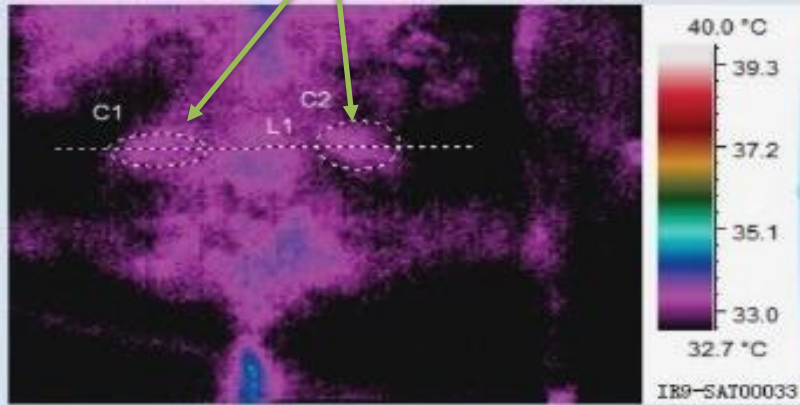
TrP = trigger point

Characteristics

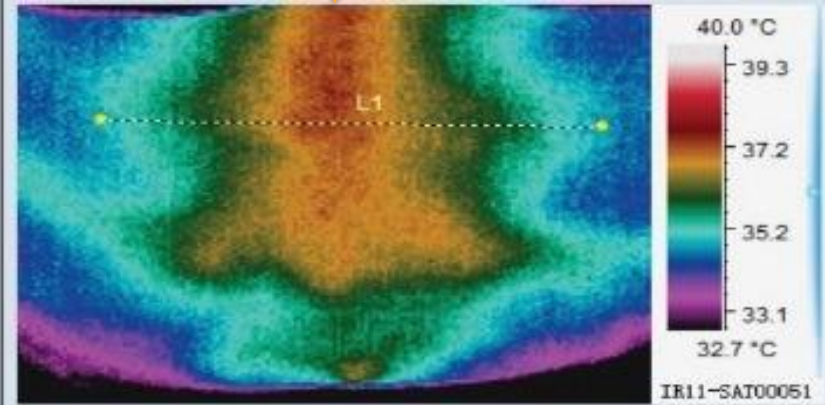
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Trigger points on thermal
imaging, before
treatment are *cooler*

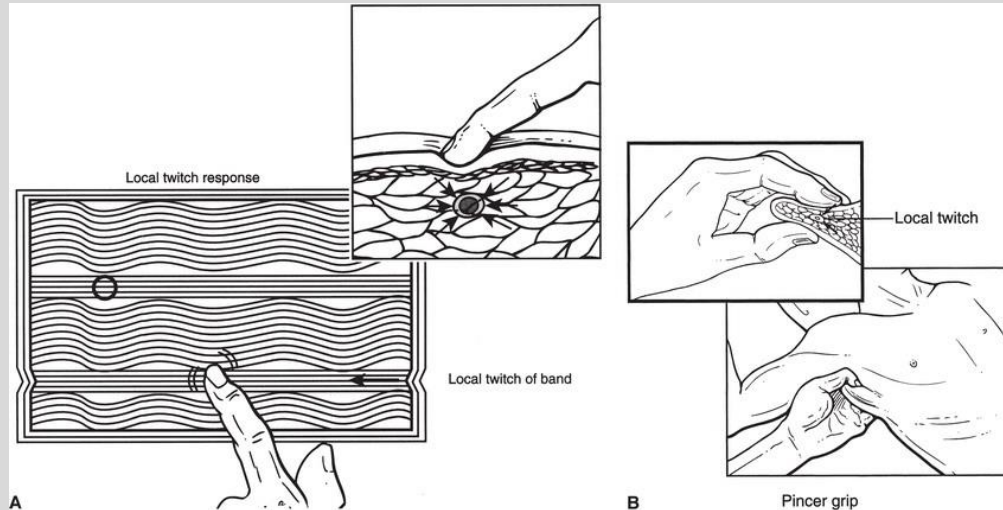


Thermal imaging 5
days after treatment:
warmer



Characteristics

- “Jump Sign” – patient pain response (wincing, withdrawal response)
- “Local Twitch Response” – transient contraction of the taut band of muscle where the trigger point is located.



Characteristics



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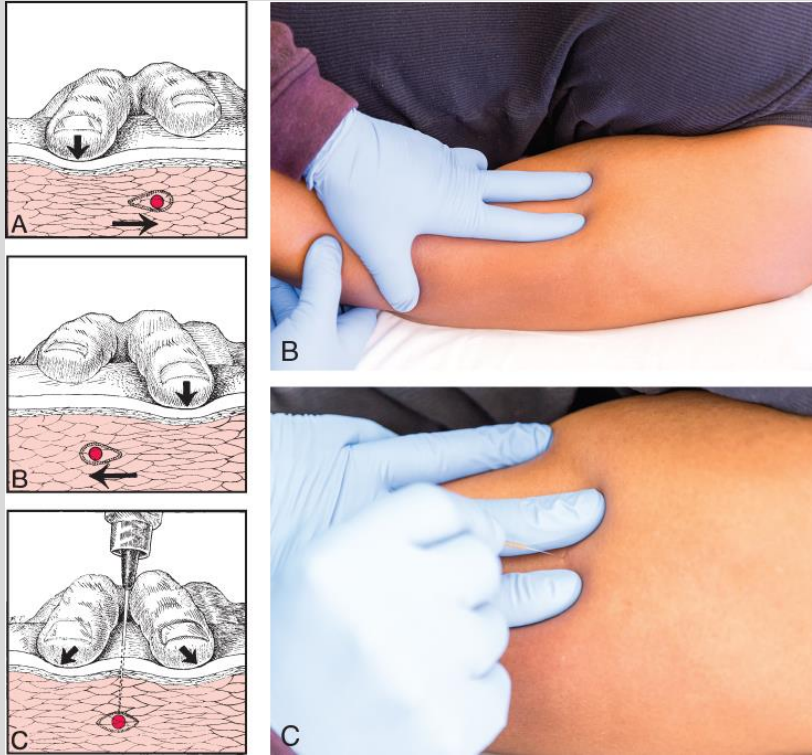
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Trigger point on ultrasound
(arrows)

Diagnosis

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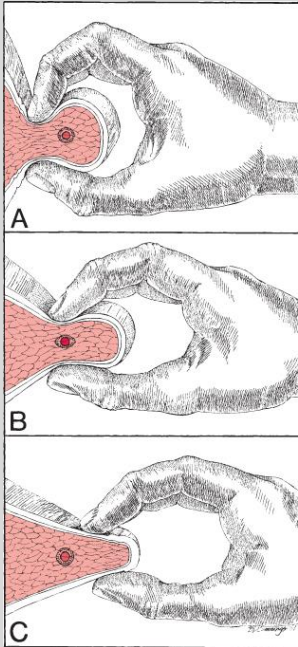


Trigger Points are palpated as small nodular or spindle-shaped thickenings within a taut band of tissue.

Diagnosis

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Palpate trigger point by applying pressure *perpendicular* to the long axis of the muscle.

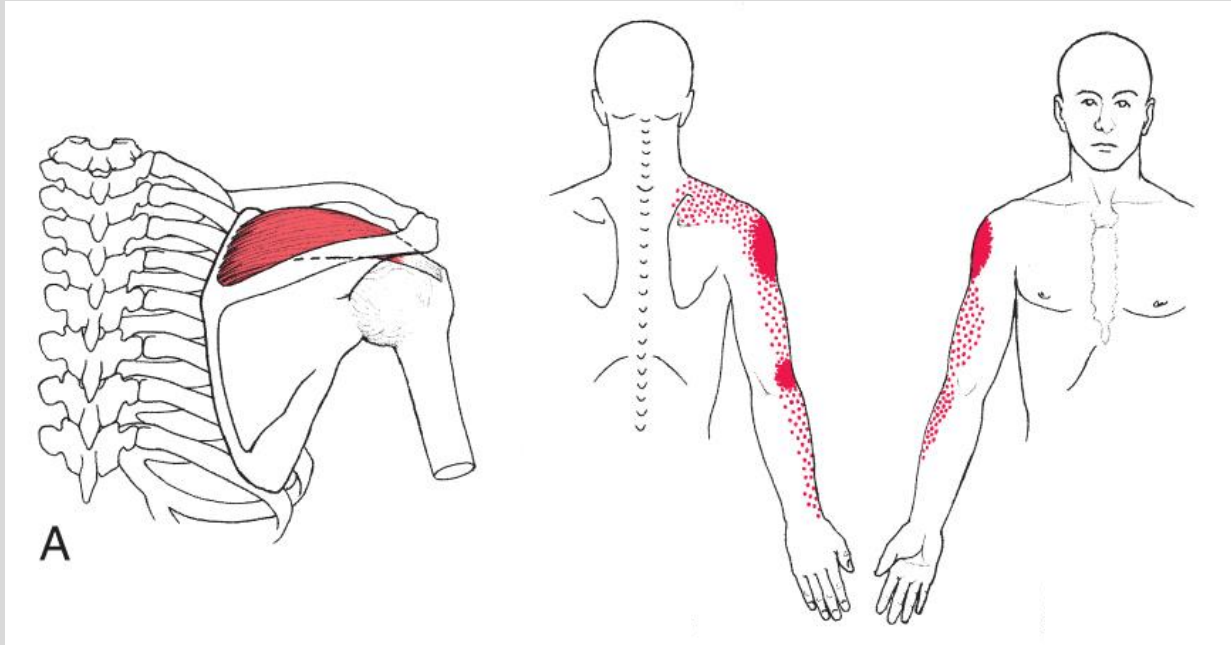
Example Trigger Point and Referral Pattern

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Supraspinatus:

“Overhead Activity Archenemy”



Diagnosis

Referral phenomena

- Sensory: pain
- Motor: weakness or stiffness
- Autonomic:
 - Vasoconstriction (blanching)
 - Cold
 - Sweating
 - Pilomotor response (goosebumps)
 - Hypersecretion

Myofascial Pain Syndrome (MPS):

The sensory, motor, and autonomic symptoms caused by myofascial trigger points, in a specific muscle or muscle group.

Causes

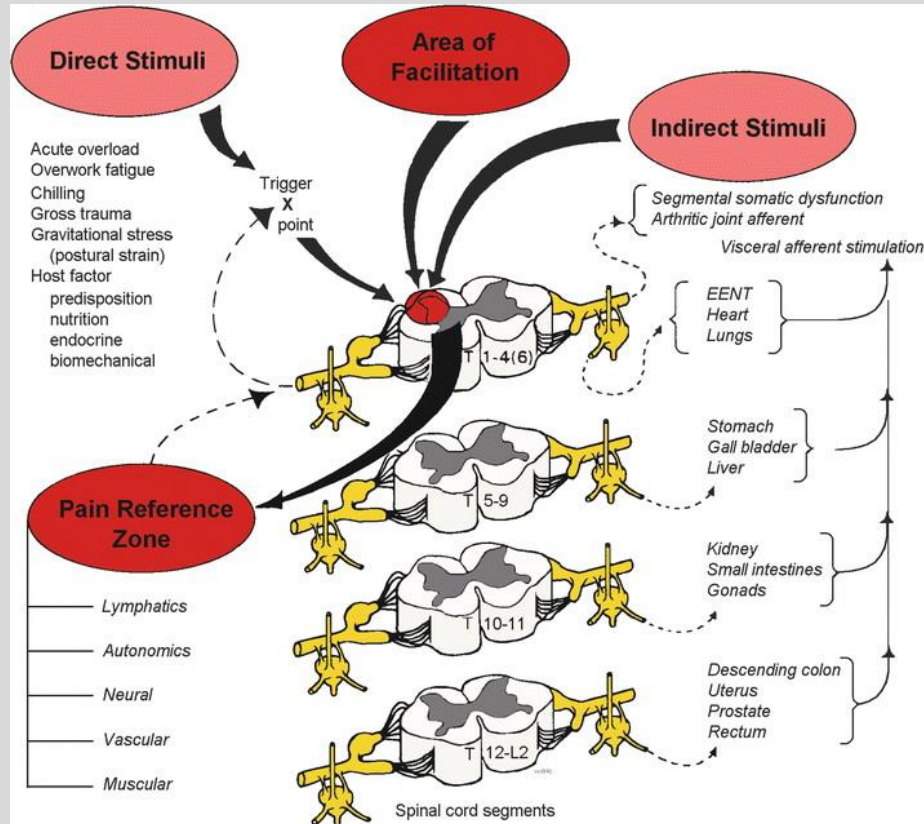
Muscle Overload Biomechanical Stress

- Types of mechanical stress
 - Postural (*ex: forward head carriage*)
 - Structural (*ex: pelvic shear*)
 - Static (*ex: acute muscle strain*)
 - Dynamic (*ex: repetitive motion*)
- Pre-disposing factors to mechanical stress include:
 - Genetic (*ex: EDS*)
 - Nutritional deficiencies
 - Metabolic imbalance (*ex: hypothyroidism*)
- Other trigger points
 - Functional myotatic unit
 - Satellite trigger points (muscles within the referral pain pattern)

Pathophysiology

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Types of Trigger Points

Active Trigger Points	Latent Trigger Points
• Refers or produces a patient's <u>familiar</u> pain (or patient's other referred symptoms) • <u>Spontaneous</u> local or referred pain	• Local or referred <u>unfamiliar</u> pain • Painful only when palpated or needled

Differential Diagnoses

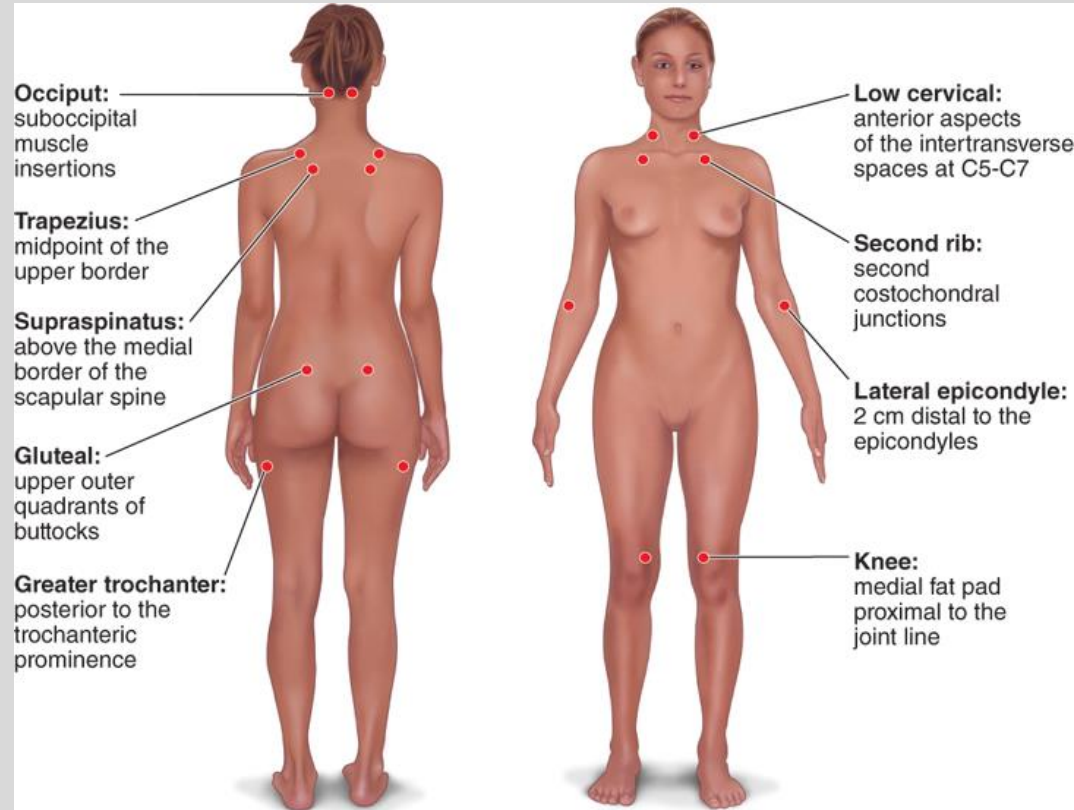
How are Trigger Points different from
Counterstrain Points/Fibromyalgia Tender Points?

- CS points are on the same continuum as trigger points
- CS points do not cause referred pain
- Fibromyalgia often includes a total body increase in pain sensitivity

Fibromyalgia Tender Points

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Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e
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Treatment

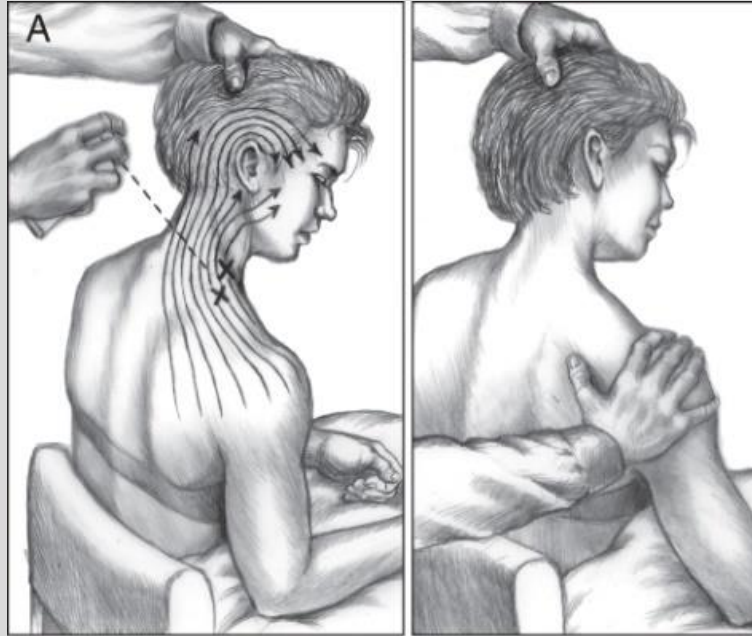
Also called “inactivation” of trigger points.

- Manual therapy
 - OMM (MET, CS, ST, HVLA, MFR, neuromuscular inhibition)
 - Spray and Stretch
- Wet needling
 - Lidocaine injection
- Dry needling (peppering)
 - No injection

Treatment

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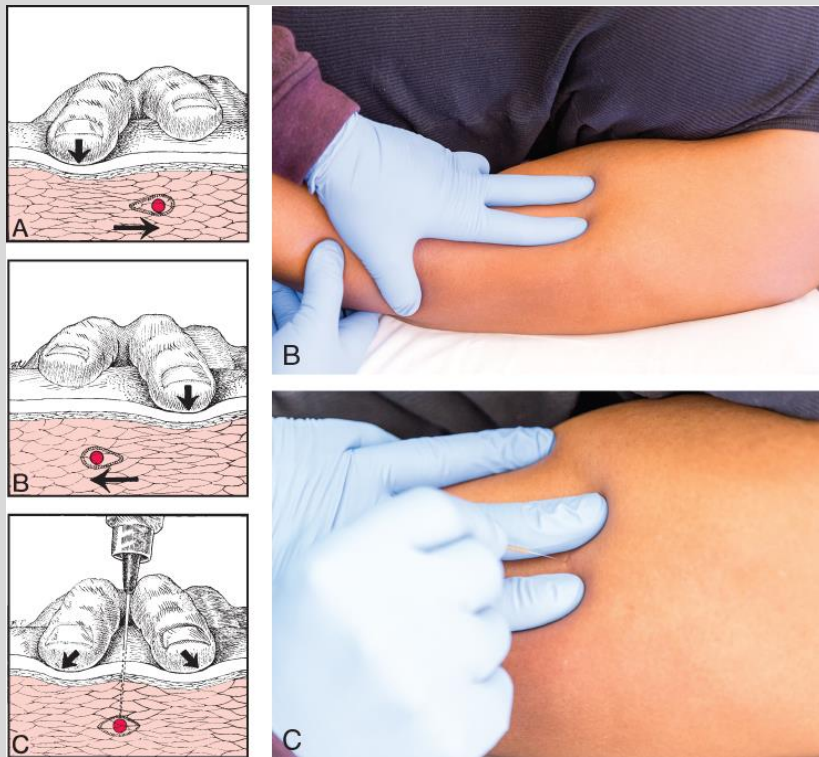
Spray

Stretch

Treatment

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Treatment

Treatment sequence:

- a. Patient lying down (psychogenic syncope)
 - b. clean site with alcohol swab
 - c. cold spray 5-6 seconds
 - d. pull skin taught between index and middle finger (this prevents the trigger point from “rolling away” as the needle tip touches it)
 - e. single skin penetration (approach at 30 degrees) with needling of multiple loci. Trigger point feels resistant like a rubber ball, and may elicit local twitch reflex and/or pain at the referral site (both are associated with good treatment outcomes)
 - f. withdraw needle and apply hemostasis (pressure)
 - g. stretch muscle (muscle energy or active patient stretch)
 - h. assess post-treatment pain
 - i. post-treatment regimen
 - i. resting and sleeping postures
 - ii. activity modification
 - iii. self-stretching exercises
 - iv. exercise prescription – stretch and strengthen
 - v. these are different for every trigger point/muscle
- Goals: improved pain and improved function

Treatment

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Treatment

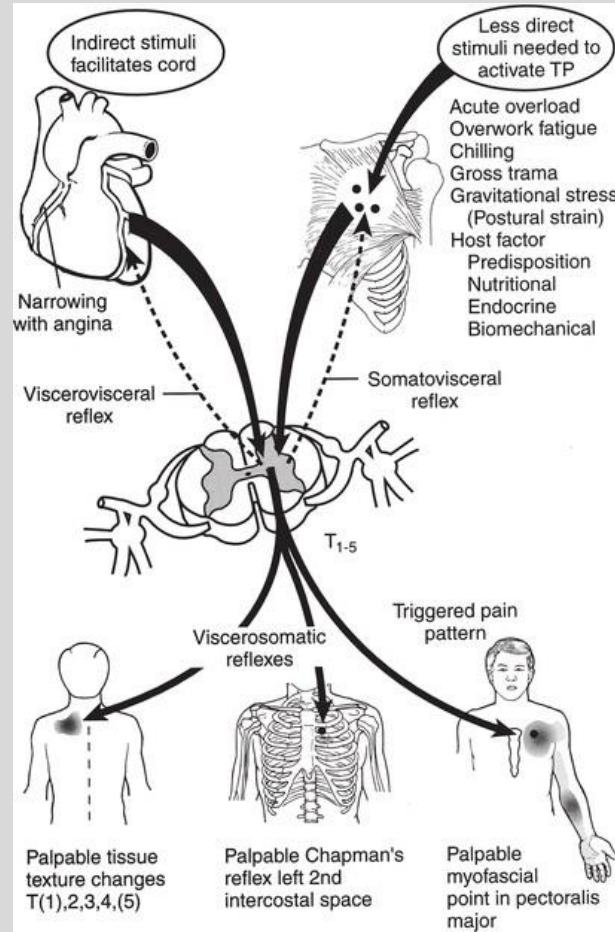
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Prevention

Examples of at-risk occupations for postural deviations:

- Truck drivers
- Construction workers
- Administrative assistants
- Dental hygienists

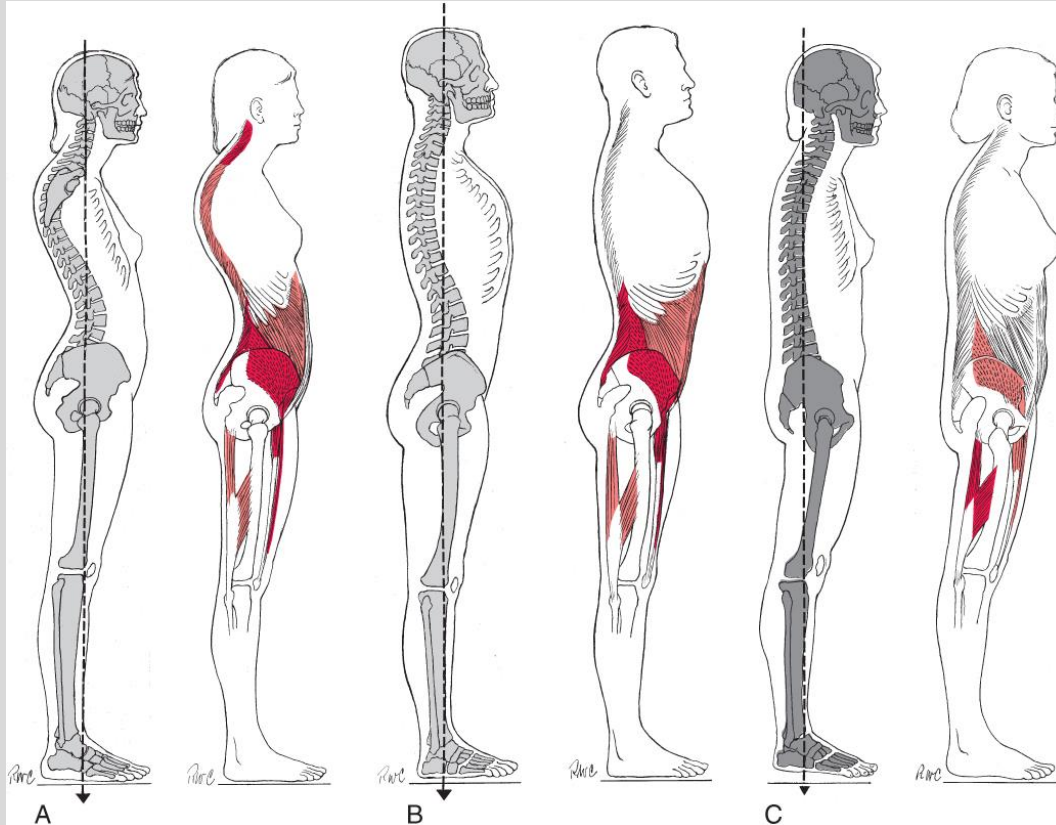
Examples of at-risk recreational activities with increased postural demand:

- Weight-lifting
- Swimming
- Cycling
- Running
- Gymnastics

Prevention of Trigger Points

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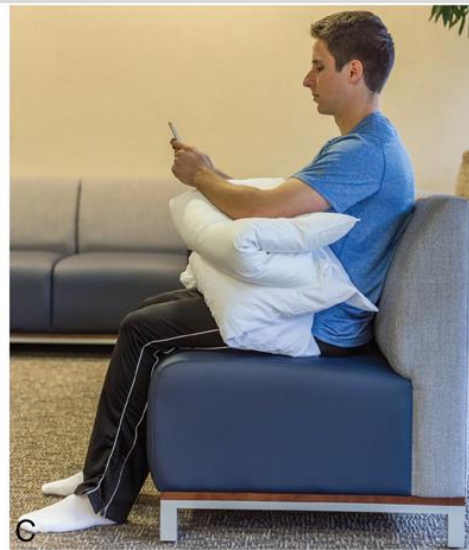
Dark red:
shortened/facilitated

Light red:
lengthened/inhibited

Prevention

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